

ASCII MASTER FLOW - PANEL 1/12: Energy-service boundary

PRIMARY RESOURCE / IMPORT -> CONVERSION -> NETWORK / STORAGE
-> DISTRIBUTION -> FINAL ENERGY -> USEFUL SERVICE -> DEVELOPMENT
TEST: delivered reliability and affordability, not capacity or connection alone

ASCII MASTER FLOW - PANEL 2/12: Accounting and units

PRIMARY != SECONDARY != FINAL != USEFUL ENERGY
MW/GW = power | MWh/GWh/BU = energy | storage needs MW + MWh
CAPACITY SHARE != GENERATION SHARE != TOTAL-ENERGY SHARE
31 Dec 2025 non-fossil capacity 51.93%; Apr-Dec 2025 generation 30.41%

ASCII MASTER FLOW - PANEL 3/12: Load and dispatch

LOAD CURVE -> BASE + INTERMEDIATE + PEAK REQUIREMENTS
ELIGIBLE CAPACITY -> security/technical constraints -> merit order -> dispatch
PLF/capacity factor = utilisation, not efficiency
FY25 energy deficit 0.1%; peak deficit 0.0%; local quality remains separate

ASCII MASTER FLOW - PANEL 4/12: Electricity chain

GENERATION -> CTU/STU -> GRID-INDIA/NLDC/RLDC/SLDC
-> DISCOM / OPEN ACCESS -> METER -> CONSUMER
REAL TIME: schedule -> deviation -> DSM -> ancillary response

ASCII MASTER FLOW - PANEL 5/12: Law and institutions

ELECTRICITY ACT, 2003 -> licensing + open access + tariff + consumer duties
CEA technical | CERC inter-state/market | SERC state/retail | APTEL appeal
DRAFT Electricity (Amendment) Bill, 2025 != enacted law at 10 Sep 2026

ASCII MASTER FLOW - PANEL 6/12: Tariff and DISCOM

TARIFF = power + network/loss + O&M/finance + return
SUBSIDY != CROSS-SUBSIDY != REGULATORY ASSET
AT&C FY25 15.04% | ACS-ARR FY25 Rs 0.06/kWh
UDAY debt reset (2015) != RDSS conditional reform (sunset 31 Mar 2028)

ASCII MASTER FLOW - PANEL 7/12: Coal system

MINE -> GRADE -> EVACUATION -> STOCK -> GENERATION -> ASH/CLOSURE
FY25 production 1,047.523 MT
FY26 imports 246.37 MT = coking 66.33 + non-coking 180.04
LINKAGE/SHAKTI != MINE AUCTION != E-AUCTION

ASCII MASTER FLOW - PANEL 8/12: Oil and gas

UPSTREAM -> MIDSTREAM -> DOWNSTREAM; PNGRB excludes production
PPAC FY26 provisional: crude dependence 88.7%; gas dependence 50.1%
LNG -> regas -> pipe -> CGD/fertiliser/power/industry
SPR Phase-I 5.33 MMT; approved Phase-II 6.5 MMT != operational stock

ASCII MASTER FLOW - PANEL 9/12: Renewable portfolio

SOLAR daytime variable | WIND seasonal | HYDRO flexible/hydrology bound
BIOMASS feedstock bound | NUCLEAR firm and institutionally bounded
AUCTION -> award -> finance -> grid -> commission -> generation
LOW BID != delivered tariff; NO FUEL COST != zero system cost

ASCII MASTER FLOW - PANEL 10/12: Integration and storage

FORECAST + TRANSMISSION/GEC + FLEXIBLE SUPPLY + DEMAND RESPONSE
+ BESS/PSP + RESERVES/MARKETS -> RELIABLE VARIABLE-RE SERVICE
CEA need: 336 GWh by 2029-30; 411 GWh by 2031-32
STATE MW + MWh + duration + cycles + efficiency

ASCII MASTER FLOW - PANEL 11/12: Instruments

RPO/RCO = renewable obligation | ESO = storage obligation
REC = 1 MWh renewable attribute | PAT ESCert = energy saving
CCTS credit = GHG performance; NO automatic fungibility
Green H2 target >=5 MMT/year by 2030; allocation != commissioning

ASCII MASTER FLOW - PANEL 12/12: Security and answer route

AVAILABILITY + ACCESS + AFFORDABILITY + RELIABILITY + SUSTAINABILITY + RESILIENCE
-> diversify source/route/contract + stocks + efficiency + finance + recovery
ANSWER: define/date/unit -> trace chain/institutions -> diagnose economics
-> test justice/environment/import risk -> verdict on delivered system cost
CROSS-LINKS: Topic 18 infrastructure | Topic 25 climate economics